

Mohammed Bougtab

AI Engineer — Data & Agentic Systems

Casablanca, Morocco +212 6 98 64 21 74 bougtab.mohammed03@gmail.com

BougtabMohammed (<https://github.com/BougtabMohammed>)

AI Engineer based in Casablanca. I design multi-agent systems, RAG solutions and the data pipelines that feed them. Engineering degree from EMSI plus a public law degree — a dual background that led me to build a budget decision-support system for a public administration.

EXPERIENCE

AI & Data Engineer — Moroccan retail group, Casablanca

Jul. 2026 — present

- Designing agentic systems — LLMs, tool calling, RAG — that take over work business teams used to do by hand.
- Integration pipelines pulling together heterogeneous sources, so agents answer from reliable, current context rather than plausible guesses.
- Systematic evaluation of agent responses: an AI system without an evaluation protocol is not production-ready, only demo-ready.
- Turning business needs directly into architecture, working alongside the teams — where half the value of an AI project is actually decided.
- Continuous improvement in production: tracking failure cases, tuning tools and prompts, trading off cost, latency and quality.

Data & AI Engineer — IT consulting firm, Rabat

Mar. — Jul. 2026

- Agentic architecture orchestrated with LangChain and LangGraph: a supervisor agent, three specialised tools, explicit decision workflows rather than a prompt chain.
- Prediction tool — Scikit-Learn and Pandas models for budget projection and trend analysis.
- RAG tool — contextual search across standards and reference frameworks: chunking, embeddings, vector store, document indexing.
- Query tool — optimised SQL against the national price reference database.
- LLM response evaluation, performance tuning and Docker containerisation for handover to the client environment.

Data Scientist — independent — Freelance engagements, Casablanca

Dec. 2025 — Jan. 2026

- Société Marocaine de Recouvrement — a payment-propensity scoring model combining response history and settlement behaviour, to prioritise collection agents' work.
- Brands & Corners (formerly Virgin Megastore) — during the conversion of an international franchise into a fully Moroccan brand: market-basket association rules with Apriori and FP-Growth.
- Brands & Corners — sales forecasting with Prophet, to anticipate demand and support commercial decisions.
- Scoping, methodology and presentation of results handled alone, directly with business stakeholders.

Back-End Developer — 6Solutions, Casablanca

Jul. — Sep. 2025

- REST APIs in Spring Boot and a relational model designed for integration rather than demonstration.
- Two-way synchronisation of products and orders with the Dolibarr ERP.
- This is where I learned to ship software that lasts — the skill that separates a deployed AI system from a successful notebook.

Data Analyst — Yola Fresh, Casablanca

Jul. — Aug. 2024

- Extraction, transformation and analysis using advanced SQL, Python and Excel.
- Power BI dashboards for tracking operational indicators.
- Performance analysis and decision-oriented insight reporting.

EDUCATION

EMSI Casablanca — Engineering degree — 5 years

2021 — 2026

Computer Science & Networks, MIAGE specialisation

Université Hassan II de Casablanca — Bachelor's degree

2023 — 2026

Public Law, International Relations and Political Science

SKILLS

AI & AGENTS	LLM · RAG · Tool calling · Systèmes multi-agents · LangChain · LangGraph · Embeddings · Vector stores · Évaluation LLM
DATA & MACHINE LEARNING	Python · Pandas · Scikit-Learn · Prophet · Apache Kafka · PySpark · NLP · SQL · PostgreSQL
CLOUD & MLOPS	Google Cloud Platform · BigQuery ML · Vertex AI · Dataflow · Docker
SOFTWARE ENGINEERING	Java · Spring Boot · APIs REST · Modélisation relationnelle · Git
DELIVERY & BUSINESS	Power BI · Analyse exploratoire · Cadrage du besoin · Communication métier · Finances publiques

CERTIFICATIONS

IBM Data Engineering Professional Certificate — IBM

The Structured Query Language (SQL) — University of Colorado Boulder

LANGUAGES

Arabic (Native) · French (C1 — advanced professional) · English (B2 — professional)